Infrared Spectroscopy and Mass Spectrometry

Answer key with worked solutions

A. IR spectroscopy

1. IR basics.

> (a) The bond **vibrates** more strongly (stretches or bends): it absorbs the energy of IR radiation at a wavenumber characteristic of that bond. (b) The region **below about 1500 $\mathbf{cm^{-1}}$**: a complex pattern unique to each compound, **compared with a database** of known spectra to confirm an identity.

2. Functional groups.

> (a) **alcohol** (O–H and C–O) (b) **C=O** with no O–H: a ketone, aldehyde or ester (c) **carboxylic acid** (very broad O–H and C=O) (d) **nitrile**, C≡N

3. Wavelengths.

> (a) $\lambda = \dfrac{1}{3300\ \text{cm}^{-1}} = 3.030 \times 10^{-4}\ \text{cm} \times \dfrac{10^4\ \mu\text{m}}{1\ \text{cm}} = \mathbf{3.03\ \mu m}$
> (b) $\lambda = \dfrac{1}{2250\ \text{cm}^{-1}} = 4.444 \times 10^{-4}\ \text{cm} = \mathbf{4.44\ \mu m}$

4. Three C3 compounds.

> **Propan-1-ol**: broad O–H at 3200–3550 cm^{-1}, **no C=O**. **Propanal**: strong C=O at about 1730 cm^{-1}, **no O–H**. **Propanoic acid**: **both** a very broad O–H (2500–3300 cm^{-1}) and a C=O.

B. Mass spectrometry

5. Butanone.

> **72**: molecular ion $[CH_3COCH_2CH_3]^+$ (C_4H_8O, 72). **57**: $CH_3CH_2CO^+$ (loss of CH_3, 72 − 15). **43**: CH_3CO^+ (loss of C_2H_5, 72 − 29). **29**: ${C_2H_5}^+$.

6. Halogens.

> 3 : 1 for M and M+2: **chlorine** (one Cl atom). 1 : 1: **bromine** (one Br atom).

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7. Bromine.

$$A_\mathrm{r} = 78.918 \times 0.5069 + 80.916 \times 0.4931$$
$$= \mathbf{79.90}$$

C. Challenge

8. The unknown.

Ethanoic acid, $\mathrm{CH_3COOH}$. Very broad 2500–3300 $\mathrm{cm^{-1}}$: **acid O–H**; 1710 $\mathrm{cm^{-1}}$: **C=O**. MS: **60** = molecular ion ($\mathrm{C_2H_4O_2}$, 60 g/mol); **45** = $\mathrm{COOH^+}$ (loss of $\mathrm{CH_3}$); **43** = $\mathrm{CH_3CO^+}$ (loss of OH, 60 − 17); **15** = $\mathrm{CH_3^+}$.

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